

RTOS

Understanding Context Switch in RTOS

Problem Statement

In an RTOS, what does the term 'context switch' refer to?

Time Spent
0 min 43 secs

Difficulty

Easy

Skills

RTOS

Score

1/1

Candidate Response

☒ A) Saving the current state of a task and loading the state of the next task to run

☐ B) Switching from user mode to kernel mode

☐ C) Transitioning from one interrupt to another

☐ D) Changing the priority of the current task

Expected Response

☒ A) Saving the current state of a task and loading the state of the next task to run

☐ B) Switching from user mode to kernel mode

☐ C) Transitioning from one interrupt to another

☐ D) Changing the priority of the current task

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Specifications used for defining timing of system calls

Problem Statement

Which of the given specifications can be used to define the timing of system calls in RTOS?

1. Task switching latency
2. Interrupt latency

Time Spent
0 min 55 secs

Difficulty

Easy

Skills

RTOS

Score

1/1

Candidate Response

☐ Only 1

☐ Only 2

☒ Both 1,2

☐ None of the given options

Expected Response

☐ Only 1

☐ Only 2

☒ Both 1,2

☐ None of the given options

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

IPC

Problem Statement

Message passing system allows processes to _____

Not applicable

Time Spent

0 min 34 secs

Difficulty

Easy

Skills

RTOS

Score

1 / 1

Candidate Response

☒ communicate with each other without sharing the same address space

☐ communicate with one another by resorting to shared data

☐ share data

☐ name the recipient or sender of the message

Expected Response

☒ communicate with each other without sharing the same address space

☐ communicate with one another by resorting to shared data

☐ share data

☐ name the recipient or sender of the message

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Scheduling

Problem Statement

The performance of Round Robin algorithm depends heavily on

Not applicable

Time Spent

0 min 25 secs

Difficulty

Easy

Skills

RTOS

Score

1 / 1

Candidate Response

☐ Size of the process

☐ I/O bursts of the process

☐ CPU bursts of the process

☒ Size of the time quantum

Expected Response

☐ Size of the process

☐ I/O bursts of the process

☐ CPU bursts of the process

☒ Size of the time quantum

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Scheduling

Problem Statement

Option to avoid deadlock situation is ____

Not applicable

Time Spent

0 min 38 secs

Difficulty

Easy

Skills

RTOS

Score

1/1

Candidate Response

☐ Disable interrupts

☐ Configure semaphore

☒ Timeout handling in process

☐ Message queues

Expected Response

☐ Disable interrupts

☐ Configure semaphore

☒ Timeout handling in process

☐ Message queues

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Scheduling

Problem Statement

Which of the following process state transitions is/are NOT possible?

Not applicable

Time Spent

3 mins 0 sec

Difficulty

Easy

Skills

RTOS

Score

1/1

Candidate Response

☐ Running to Ready

☒ Waiting to Running

☒ Ready to Waiting

☐ Running to Terminated

Expected Response

☐ Running to Ready

☒ Waiting to Running

☒ Ready to Waiting

☐ Running to Terminated

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

ISR

Problem Statement

When the process is returned after an interrupt service, which of the following should be loaded again?

Time Spent Difficulty

0 min 51 secs

Easy

Skills

RTOS

Score

0 / 1

Not applicable

Candidate Response

☒ Register contents

☐ Condition codes

☒ Stack contents

☒ Return addresses

Expected Response

☒ Register contents

☐ Condition codes

☒ Stack contents

☐ Return addresses

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

CPU Load

Problem Statement

CPU utilization % is calculated by using which formula?

Time Spent Difficulty

0 min 22 secs

Medium

Skills

Embedded

Score

2 / 2

Not applicable

Candidate Response

☐ (Completion Time – Arrival Time)

☐ (Time is started executing – Arrival Time)

☐ (Number of processes) / (Max of completion time – Time at which first process comes into CPU)

☒ (Max of Completion time – Idle Time)/(Max of Completion Time)

Expected Response

☐ (Completion Time – Arrival Time)

☐ (Time is started executing – Arrival Time)

☐ (Number of processes) / (Max of completion time – Time at which first process comes into CPU)

☒ (Max of Completion time – Idle Time)/(Max of Completion Time)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Properties of different types of threads

Problem Statement

Which of these threads will run only when no other thread is ready to execute with respect to an RTOS application?

Time Spent

1 min 23 secs

Difficulty

Medium

Skills

RTOS

Score

2 / 2

Candidate Response

☐ Interrupt Service Routine

☒ Idle

☐ Tasks

☐ Preemptive Tasks initializer

Expected Response

☐ Interrupt Service Routine

☒ Idle

☐ Tasks

☐ Preemptive Tasks initializer

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Working on an RTOS based application

Problem Statement

You are working on an RTOS based application where you are using semaphores for resource management. When doing so, a task gets blocked until a resource required by it is unavailable.

Which of these methods should you execute to fix this issue?

Time Spent

3 mins 19 secs

Difficulty

Medium

Skills

RTOS

Score

2 / 2

Candidate Response

☒ sem_post()

☐ sem_wait()

☐ sem_release()

☐ sem_put()

Expected Response

☒ sem_post()

☐ sem_wait()

☐ sem_release()

☐ sem_put()

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Ordering affected by Scheduling policy

Problem Statement

Which of the following scenarios will impact process ordering?

1. When a process is a running thread and it becomes a blocked thread.
2. When a process is a running thread and it becomes a preempted thread.

Time Spent

1 min 54 secs

Difficulty

Medium

Skills

RTOS

Score

0 / 2

Candidate Response

☐ Only 1

☒ Only 2

☐ Both 1 and 2

☐ Neither 1 nor 2

Expected Response

☐ Only 1

☐ Only 2

☒ Both 1 and 2

☐ Neither 1 nor 2

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

RTOS Message Queue

Problem Statement

You have created a message queue in RTOS of size zero.

What happens when a task tries to send a message to the queue and there is no task waiting for receiving messages at this moment?

Time Spent

1 min 46 secs

Difficulty

Medium

Skills

RTOS

Score

2 / 2

Candidate Response

☒ The producer task will receive an error code saying that the queue is full

☐ The producer task will fail to send a message

☐ The producer task will successfully send a message

☐ None of the given options

Expected Response

☒ The producer task will receive an error code saying that the queue is full

☐ The producer task will fail to send a message

☐ The producer task will successfully send a message

☐ None of the given options

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Periodic Task Execution in RTOS

Problem Statement

When using a Real-Time Operating System (RTOS), which feature allows tasks to be executed periodically?

Time Spent
0 min 51 secs

Difficulty
Medium

Skills

RTOS

Score

2 / 2

Candidate Response

- ☐ Priority Scheduling
- ☐ Semaphore
- ☒ Timer
- ☐ Message Queue

Expected Response

- ☐ Priority Scheduling
- ☐ Semaphore
- ☒ Timer
- ☐ Message Queue

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

RTOS Task Scheduling

Problem Statement

In the context of a Real-Time Operating System (RTOS), what is the primary function of a task scheduler?

Time Spent
0 min 21 secs

Difficulty
Medium

Skills

RTOS (Real-Time Operating Systems)

Score

2 / 2

Candidate Response

- ☐ To manage the memory allocation of running tasks.
- ☐ To control the synchronization between tasks and interrupts.
- ☒ To determine the order in which tasks are executed based on their priorities.
- ☐ To provide a communication mechanism between different tasks.

Expected Response

- ☐ To manage the memory allocation of running tasks.
- ☐ To control the synchronization between tasks and interrupts.
- ☒ To determine the order in which tasks are executed based on their priorities.
- ☐ To provide a communication mechanism between different tasks.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Understanding Priority Inversion in RTOS

Problem Statement

In a Real-Time Operating System (RTOS), what is Priority Inversion?

Time Spent

0 min 15 secs

Difficulty

Medium

Skills

RTOS (Real-Time Operating Systems)

Score

0 / 2

Candidate Response

- ☐ 1. When a higher priority task is unable to run because a lower priority task has not yet completed.
- ☐ 2. When tasks share priorities and execute in a round-robin fashion.
- ☒ 3. When a lower priority task preempts a higher priority task.
- ☐ 4. When two or more tasks are stuck waiting for each other to release resources.

Expected Response

- ☒ 1. When a higher priority task is unable to run because a lower priority task has not yet completed.
- ☐ 2. When tasks share priorities and execute in a round-robin fashion.
- ☐ 3. When a lower priority task preempts a higher priority task.
- ☐ 4. When two or more tasks are stuck waiting for each other to release resources.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

RTOS Task Issues

Problem Statement

In an RTOS-based system, why might tasks not run in the expected order despite being correctly prioritized?

Time Spent

1 min 25 secs

Difficulty

Medium

Skills

Embedded Systems

Score

0 / 2

Candidate Response

- ☐ Insufficient stack size
- ☐ Task starvation
- ☐ Incorrect use of semaphores
- ☒ Task deadlock

Expected Response

- ☐ Insufficient stack size
- ☐ Task starvation
- ☒ Incorrect use of semaphores
- ☐ Task deadlock

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Depict the outcome of an event

Problem Statement

A thread in a multithreading environment already has a non-recursive mutex associated with it. However, the thread which has already locked the mutex tries to lock the mutex again.

Which of these statements correctly depict the outcome of such an event?

Time Spent
3 mins 56 secs

Difficulty
Hard

Skills
Operating Systems

Score
0 / 2

Candidate Response

- ☒ The thread gets locked again but a count is associated with the mutex. The thread has to be unlocked as many times as it has been locked by the mutex.
- ☐ The thread enters into the waiting list for the same mutex and the waiting time doesn't end because the thread is already locked by the same mutex. This results in a deadlock.
- ☐ The requesting thread is moved from the running list of processor to the waiting list of the synchronization primitive. Once the mutex is available, the thread with the highest priority according to the scheduling algorithms in the list gets locked by the mutex.
- ☐ The thread calls another mutex and gets associated with it too and thereby acquires an additional resource.

Expected Response

- ☐ The thread gets locked again but a count is associated with the mutex. The thread has to be unlocked as many times as it has been locked by the mutex.
- ☒ The thread enters into the waiting list for the same mutex and the waiting time doesn't end because the thread is already locked by the same mutex. This results in a deadlock.
- ☐ The requesting thread is moved from the running list of processor to the waiting list of the synchronization primitive. Once the mutex is available, the thread with the highest priority according to the scheduling algorithms in the list gets locked by the mutex.
- ☐ The thread calls another mutex and gets associated with it too and thereby acquires an additional resource.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Advantages of using Scheduling tasks

Problem Statement

Three RTOS tasks t1, t2, and t3 are scheduled as follows - t1 runs until it has completed its work, then terminates. Then the next task t2 runs similarly and so forth until all tasks are completed.

What is the advantage of scheduling tasks in this manner?

Time Spent
0 min 38 secs

Difficulty
Hard

Skills
RTOS

Score
3 / 3

Candidate Response

- ☐ Each task's allocation of time is totally affected by all the other tasks.
- ☒ None of the tasks is pre-empted and each task completes in a definite time.
- ☐ The system will be very deterministic
- ☐ None of the given options

Expected Response

- ☐ Each task's allocation of time is totally affected by all the other tasks.
- ☒ None of the tasks is pre-empted and each task completes in a definite time.
- ☐ The system will be very deterministic
- ☐ None of the given options

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Carrying out Interprocess Communication

Problem Statement

Column A consists of multiple approaches of carrying out Inter-process Communication whereas Column B contains their descriptions. Match the columns and mark the appropriate options.

Time Spent

2 mins 5 secs

Difficulty

Hard

Skills

RTOS

Score

0 / 3

Column A

1.Socket

2.Semaphore

Expand Question ▾

Candidate Response

☐ 1-d ☐ 2-b ☐ 3-c ☐ 4-a
☐ 1-a ☐ 2-d ☐ 3-c ☐ 4-b
☒ 1-a ☐ 2-b ☐ 3-c ☐ 4-d
☐ 1-d ☐ 2-a ☐ 3-c ☐ 4-b

Expected Response

☒ 1-d ☐ 2-b ☐ 3-c ☐ 4-a
☐ 1-a ☐ 2-d ☐ 3-c ☐ 4-b
☐ 1-a ☐ 2-b ☐ 3-c ☐ 4-d
☐ 1-d ☐ 2-a ☐ 3-c ☐ 4-b

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Unique priorities being viable option for threads in RTOS

Problem Statement

In which of the given scenarios is the assignment of unique priorities a viable option for threads in RTOS?

Time Spent

3 mins 5 secs

Difficulty

Hard

Skills

RTOS

Score

0 / 3

Candidate Response

☒ when the number of context switches need to be minimal
☐ latency is more important than throughput for the system
☐ throughput is more important than latency
☐ when the system does not need to be aware of the priority assigned to each thread

Expected Response

☐ when the number of context switches need to be minimal
☒ latency is more important than throughput for the system
☐ throughput is more important than latency
☐ when the system does not need to be aware of the priority assigned to each thread

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Semaphores properties in RTOS

Problem Statement

You have two RTOS tasks t1 and t2. Both t1 and t2 need access to a shared resource, but t1 has already acquired the semaphore. The task t1 has not released the semaphore.

Will t2 be able to access the shared resource?

Note: Assume that t2 has a higher priority than t1.

Time Spent Difficulty

1 min 56 secs

Hard

Skills

RTOS

Score

3 / 3

Candidate Response

☐ t2 will be able to access the shared resource because of higher priority.

☐ t2 will not be able to access the shared resource until t1 has released the semaphore.

☐ t2 will be in a wait state until semaphore is released by t1.

☒ Both Choice 2 and 3

Expected Response

☐ t2 will be able to access the shared resource because of higher priority.

☐ t2 will not be able to access the shared resource until t1 has released the semaphore.

☐ t2 will be in a wait state until semaphore is released by t1.

☒ Both Choice 2 and 3

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Working on an RTOS based application for communication

Problem Statement

The threads which you are using while working on an RTOS based application communicate with each other by sending/receiving messages. In the given scenario, you want to centralize a specific functionality into a single task so that the threads can send messages to this task for processing.

Which of these mechanisms will you use in the given scenario?

Time Spent Difficulty

1 min 7 secs

Medium

Skills

RTOS

Score

1 / 1

Candidate Response

☐ Scheduler

☒ Message Queue

☐ Semaphore

☐ Mutexes

Expected Response

☐ Scheduler

☒ Message Queue

☐ Semaphore

☐ Mutexes

RTOS and Task Scheduling

Problem Statement

What will be the outcome of the following code when executed on a microcontroller with an RTOS?

```
void Task1(void *pvParameters) {
    while(1) {
        // Task 1 code
    }
}

void Task2(void *pvParameters) {
    while(1) {
        // Task 2 code
    }
}

int main(void) {
    xTaskCreate(Task1, "Task 1", 1000, NULL, 1, NULL);
    xTaskCreate(Task2, "Task 2", 1000, NULL, 1, NULL);
    vTaskStartScheduler();
    for (;;) ;
}
```

Time Spent

1 min 46 secs

Difficulty

Medium

Skills

Microcontrollers

Score

2 / 2

Candidate Response

- ☐ Task 1 will run continuously, and Task 2 will never run.
- ☐ Task 2 will run continuously, and Task 1 will never run.
- ☒ Both Task 1 and Task 2 will run alternately based on time-slicing.
- ☐ Both Task 1 and Task 2 will start together and run simultaneously.

Expected Response

- ☐ Task 1 will run continuously, and Task 2 will never run.
- ☐ Task 2 will run continuously, and Task 1 will never run.
- ☒ Both Task 1 and Task 2 will run alternately based on time-slicing.
- ☐ Both Task 1 and Task 2 will start together and run simultaneously.

Deadline Analysis

Problem Statement

In a real-time embedded system, you observe that the system occasionally misses deadlines under high-load conditions.

What is the most probable cause?

Time Spent

1 min 53 secs

Difficulty

Hard

Skills

Embedded Systems

Score

0 / 3

Candidate Response

- ☒ Insufficient RAM
- ☐ Task priority inversion
- ☐ Hardware design
- ☐ Power supply fluctuations

Expected Response

- ☐ Insufficient RAM
- ☒ Task priority inversion
- ☐ Hardware design
- ☐ Power supply fluctuations

Working with threads

Problem Statement

Which of the following is/are probable value(s) of the variable **a** in the given code snippet assuming that there are three other threads reading the value of **a**?

```
int a = 9;
a++;
```

Time Spent 1 min 25 secs
Difficulty Medium

Skills
Embedded C

Score
2 / 2

Candidate Response

☐ 10

☐ 11

☐ 9

☒ Both Choice 1 and 3

Expected Response

☐ 10

☐ 11

☐ 9

☒ Both Choice 1 and 3

RTOS concepts

Problem Statement

When two or more Threads attempt to access the same resource at the same time and try to change it at the same time, it results in a _____.

Time Spent 0 min 18 secs
Difficulty Easy

Skills
RTOS (Real-Time Operating Systems)

Score
1 / 1

Candidate Response

☒ race condition

☐ deadlock

☐ priority inversion

☐ context switching

Expected Response

☒ race condition

☐ deadlock

☐ priority inversion

☐ context switching

123456789101112131415161718192021222324252627

Task execution in Embedded Systems

Problem Statement

In the following C code snippet for an embedded system, what will be the output if the system is running in real-time with an RTOS, considering task preemption?

```
#include
#include
#include

void vTaskCode(void *pvParameters) {
    int x = 0;
    while (x < 10) {
        x++;
        printf("Value of x: %d\n", x);
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}

int main() {
    xTaskCreate(vTaskCode, "Task 1", 1000, NULL, 1, NULL);
    vTaskStartScheduler();
    for (;;)
        return 0;
}
```

Time Spent

2 mins 28 secs

Difficulty

Medium

Skills

C Programming

Score

2 / 2

Candidate Response

☐ A predefined response from the RTOS scheduler

☒ The value of the variable 'x' after the task completes

☐ An error due to task preemption

☐ Task switching with unpredictable output

Expected Response

☐ A predefined response from the RTOS scheduler

☒ The value of the variable 'x' after the task completes

☐ An error due to task preemption

☐ Task switching with unpredictable output

Embedded Systems

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

8051 Program Execution

Problem Statement

What happens when the following 8051 program is executed?

```
#include <reg51.h>
void main( )
{
    for ( ; ; )
    {
        P3=0 x 55;
        P3=0 x AA;
    }
}
```

Time Spent Difficulty

1 min 7 secs

Medium

Skills

Embedded C

Score

2 / 2

Candidate Response

☒ It toggles all bits of P3 continuously.

☐ The program will not get compiled.

☐ A syntax error is thrown.

☐ None of these

Expected Response

☒ It toggles all bits of P3 continuously.

☐ The program will not get compiled.

☐ A syntax error is thrown.

☐ None of these

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Executing a microcontroller program

Problem Statement

For a microcontroller program, you want to use variables that can be retained for use through the program in RAM locations.

Which of these variables would you use if the locations are not reallocated during program execution?

Time Spent Difficulty

1 min 16 secs

Medium

Skills

Embedded C

Score

0 / 2

Candidate Response

☐ Static variables

☐ Static variables declared within functions

☒ Static variables declared outside functions

☐ Automatic variables

Expected Response

☒ Static variables

☐ Static variables declared within functions

☐ Static variables declared outside functions

☐ Automatic variables

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Sending and receiving characters over UART

Problem Statement

In the given code, the transmitData() function sends a single character over UART and the receiveData() function receives a character over UART.

```
#include <reg51.h>
void configureUART() {
    // UART configuration code
}
void transmitData(char data) {
    // Transmit data code
}
void receiveData() {
    // Receive data code
}
void main() {
    configureUART();
    transmitData('A');
    char receivedData = receiveData();
    XXX
    while(1) {
        // Infinite loop
    }
}
```

Time Spent Difficulty

1 min 56 secs

Medium

Skills

Embedded C

Score

2 / 2

Which of the following statements should be inserted at the marked location **XXX** to receive a character and check if it matches the transmitted character A?

Candidate Response

☒ if (receivedData == 'A') { }

☐ while (receivedData != 'A') { }

☐ if (receivedData = 'A') { }

☐ while (receivedData = 'A') { }

Expected Response

☒ if (receivedData == 'A') { }

☐ while (receivedData != 'A') { }

☐ if (receivedData = 'A') { }

☐ while (receivedData = 'A') { }

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Declaring variables

Problem Statement

You have declared a variable **x** as follows in your 8051 embedded C program.

```
sbit x=P2^2;
```

Time Spent Difficulty

0 min 45 secs

Medium

Skills

Embedded C

Score

2 / 2

Which of the following statements is **TRUE** about **x**?

1. x can be used in place of P2.2.

2. It helps reduce the increasing complexity of the embedded program.

Candidate Response

☐ Only 1

☐ Only 2

☒ Both 1 and 2

☐ Neither 1 nor 2

Expected Response

☐ Only 1

☐ Only 2

☒ Both 1 and 2

☐ Neither 1 nor 2

Blinking an LED

Problem Statement

```
#include "gpio_cyt2b1_176_lqfp.h"

cy_stc_gpio_pin_config_t user_led_port_pin_cfg =
{
    .outVal = 0u1, /* Pin output state */
    .driveMode = CY_GPIO_DM_STRONG_IN_OFF, /* Drive mode */
    .hsiom = CY_LED0_PIN_MUX, /* HSIOM selection */
    .intEdge = 0u1, /* Interrupt Edge type */
    .intMask = 0u1, /* Interrupt enable mask */
    .vtrip = 0u1, /* Input buffer voltage trip type */
    .slewRate = 0u1, /* Output buffer slew rate */
    .driveSel = 0u1, /* Drive strength */
};

int main(void)
{
    SystemInit();
    __enable_irq();

    user_led_port_pin_cfg.hsiom = CY_LED0_PIN_MUX;
    Cy_GPIO_Pin_Init(CY_LED0_PORT, CY_LED0_PIN, XXX);

    for(;;)
    {
        // Wait 0.05 [s]
        Cy_SysTick_DelayInUs(50000u1);
        Cy_GPIO_Inv(CY_LED0_PORT, CY_LED0_PIN);
    }
}
```

The given code snippet toggles the state of the LED pin.

What should be insert in place of **XXX** for the correct configuration?

Time Spent Difficulty

1 min 27 secs

Medium

Skills

Embedded C

Score

2 / 2

Candidate Response

- ☐ CY_LED0_PIN
- ☐ CY_LED0_PORT
- ☒ &user_led_port_pin_cfg
- ☐ delay(0.05);

Expected Response

- ☐ CY_LED0_PIN
- ☐ CY_LED0_PORT
- ☒ &user_led_port_pin_cfg
- ☐ delay(0.05);

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Exception Classification

Problem Statement

Given below is a list of Exceptions that Embedded Programmers may encounter. Classify the below exceptions as TRAP (T), FAULT (F) or ABORT (A).

Time Spent
0 min 46 secs

Difficulty
Medium

- 1.Breakpoint Exception
- 2.Overflow Exception
- 3.Invalid Opcode Exception
- 4.Double Fault Exception

Skills
Embedded C

Score
3 / 3

Candidate Response

1-F
☐ 2-T
3-F
4-F

1-F
☐ 2-T
3-F
4-A

1-T
☒ 2-T
3-F
4-A

1-T
☐ 2-T
3-A
4-F

Expected Response

1-F
☐ 2-T
3-F
4-F

1-F
☐ 2-T
3-F
4-A

1-T
☒ 2-T
3-F
4-A

1-T
☐ 2-T
3-A
4-F

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

CPU has a non uniform delay pipeline

Problem Statement

A CPU has a non uniform delay pipeline with 3 stages. The stages have delays of 3ns, 2ns and 1ns, respectively.

What is the cycle time for this CPU if a buffer delay of 4ns is introduced?

Time Spent
1 min 33 secs

Difficulty
Medium

Skills
Computer Architecture

Score
0 / 2

Candidate Response

☐ 3ns
☐ 7ns
☐ 5ns
☒ 10ns

Expected Response

☐ 3ns
☒ 7ns
☐ 5ns
☐ 10ns

- 123456789101112131415161718192021222324252627

Types of RAM

Problem Statement

In which of these types of RAM is the data transferred twice during the positive edge and the negative edge of a clock cycle?

Time Spent1 min 30 secsDifficultyMediumSkillsComputer ArchitectureScore0 / 2

Candidate Response

☐ DDR SDRAM

☐ Synchronous DRAM

☐ Asynchronous DRAM

☐ Both Choice 1,2

☒ Both Choice 1,3

Expected Response

☒ DDR SDRAM

☐ Synchronous DRAM

☐ Asynchronous DRAM

☐ Both Choice 1,2

☐ Both Choice 1,3

- 123456789101112131415161718192021222324252627

External factor without which memory cannot work

Problem Statement

For the proper working of an SD-RAM module, there is a certain external factor that must be provided without which the memory cannot work.

Which of these is one such factor?

Time Spent0 min 55 secsDifficultyMediumSkillsComputer ArchitectureScore2 / 2

Candidate Response

☐ External cooling

☐ External latency regulator

☒ External clock

☐ None of the given options

Expected Response

☐ External cooling

☐ External latency regulator

☒ External clock

☐ None of the given options

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Pipeline

Problem Statement

A processor employs a technique, in which a number of processing elements work in parallel such that one instruction does the same operation to multiple elements of data, thereby getting more work per clock through the CPU. What is this technique called?

Time Spent

1 min 0 sec

Difficulty

Medium

Skills

Computer Architecture

Score

2 / 2

Candidate Response

☒ SIMD

☐ VLIW

☐ MAC

☐ MUL

Expected Response

☒ SIMD

☐ VLIW

☐ MAC

☐ MUL

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

DSP

Problem Statement

The radar system must track objects in a range from 0 to infinity, but needs to use only a small subset of the range for target acquisition

and identification. What type of processor is suitable?

Time Spent

0 min 45 secs

Difficulty

Medium

Skills

Embedded Systems

Score

2 / 2

Candidate Response

☐ Fixed-point DSP

☒ Floating-point DSP

☐ 8051 micro-controller

☐ ARM Cortex M0+

Expected Response

☐ Fixed-point DSP

☒ Floating-point DSP

☐ 8051 micro-controller

☐ ARM Cortex M0+

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

SPI Interface Issues

Problem Statement

What is a common issue when interfacing an SPI device with different voltage levels?

Time Spent
0 min 55 secs

Difficulty
Medium

Skills

Embedded Systems

Score
0 / 2

Candidate Response

☐ Signal attenuation

☒ Clock synchronization

☐ Data corruption

☐ GPIO pin damage

Expected Response

☐ Signal attenuation

☐ Clock synchronization

☒ Data corruption

☐ GPIO pin damage

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Microcontroller I2C Communication

Problem Statement

When programming a microcontroller to interact with a temperature sensor via I2C, which of the following operations is essential for initiating data transfer?

Time Spent
0 min 57 secs

Difficulty
Medium

Skills

Embedded Systems

Score
0 / 2

```
// Assume TWCR, TWEN, and TWSTA are macros for microcontroller-specific registers and bits
void initI2C() {
    // Enable the I2C interface
    TWCR = (1 << TWEN);
}

void startI2C() {
    // Send start condition
    TWCR = (1 << TWSTA) | (1 << TWEN);
    // Wait for the start condition to be transmitted
    while (!(TWCR & (1 << TWSTA)));
}
```

Candidate Response

☐ Sending a start condition

☐ Setting the clock frequency

☒ Reading the status register

☐ Writing to the data register

Expected Response

☒ Sending a start condition

☐ Setting the clock frequency

☐ Reading the status register

☐ Writing to the data register

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

ADC Sampling Issues

Problem Statement

When dealing with ADC (Analog-to-Digital Converter) in embedded systems, which issue often arises from improper sampling rates?

Time Spent
0 min 31 secs

Difficulty
Medium

Skills

Embedded Systems

Score
2 / 2

Candidate Response

☐ Quantization noise

☒ Aliasing

☐ Jitter

☐ Thermal noise

Expected Response

☐ Quantization noise

☒ Aliasing

☐ Jitter

☐ Thermal noise

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Memory Risk Assessment

Problem Statement

What is a significant risk of using dynamic memory allocation in safety-critical embedded systems?

Time Spent
1 min 0 sec

Difficulty
Medium

Skills

Embedded Systems

Score
2 / 2

Candidate Response

☒ Fragmentation

☐ Increased latency

☐ Memory leaks

☐ Task starvation

Expected Response

☒ Fragmentation

☐ Increased latency

☐ Memory leaks

☐ Task starvation

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

PWM Jitter Causes

Problem Statement

Why might an embedded system using PWM (Pulse Width Modulation) for motor control experience jitter in motor speed?

Time Spent
1 min 1 sec

Difficulty
Medium

Skills

Embedded Systems

Score
2 / 2

Candidate Response

☐ High-frequency switching noise

☒ Insufficient PWM resolution

☐ Poorly designed PID controller

☐ Ineffective cooling

Expected Response

☐ High-frequency switching noise

☒ Insufficient PWM resolution

☐ Poorly designed PID controller

☐ Ineffective cooling

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Effective Debugging Methods

Problem Statement

Which method is most effective for debugging a microcontroller that occasionally crashes due to a hard-to-reproduce bug?

Time Spent Difficulty

0 min 36 secs

Hard

Skills

Embedded Systems

Score

0 / 3

Candidate Response

☐ Adding more print statements

☐ Using a hardware debugger with breakpoints

☐ Increasing the clock speed

☒ Using a watchdog timer

Expected Response

☐ Adding more print statements

☒ Using a hardware debugger with breakpoints

☐ Increasing the clock speed

☐ Using a watchdog timer

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Power Consequences

Problem Statement

What is a typical consequence of poor power management in battery-operated embedded systems?

Time Spent Difficulty

1 min 12 secs

Easy

Skills

Embedded Systems

Score

0 / 1

Candidate Response

☒ Reduced system performance

☐ Increased thermal noise

☐ Shortened battery life

☐ Higher electromagnetic interference

Expected Response

☐ Reduced system performance

☐ Increased thermal noise

☒ Shortened battery life

☐ Higher electromagnetic interference

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Interrupt Importance

Problem Statement

Why is it important to disable interrupts when accessing shared data in a real-time system?

Time Spent Difficulty

0 min 50 secs

Easy

Skills

Embedded Systems

Score

1/1

Candidate Response

☐ To improve execution speed

☒ To avoid data inconsistency

☐ To save power

☐ To prevent memory leaks

Expected Response

☐ To improve execution speed

☒ To avoid data inconsistency

☐ To save power

☐ To prevent memory leaks

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Interrupt is triggered

Problem Statement

Which of these types of interrupts are triggered due to any exception that has been caused by the program itself?

Time Spent Difficulty

0 min 30 secs

Easy

Skills

Computer Architecture

Score

1/1

Candidate Response

☐ External Interrupts

☒ Internal Interrupts

☐ Hardware interrupts

☐ Both Choice 2 and 3

Expected Response

☐ External Interrupts

☒ Internal Interrupts

☐ Hardware interrupts

☐ Both Choice 2 and 3

Power Management in Microcontrollers

Problem Statement

You are designing a power-efficient embedded system and need to ensure the microcontroller only consumes power when necessary. What is the most appropriate technique to achieve this?

Time Spent 0 min 32 secs
Difficulty Medium

Skills

Microcontrollers

Score

2 / 2

Candidate Response

- ☐ A) Use a high-power microcontroller that completes tasks quickly to minimize active time.
- ☒ B) Implement deep sleep modes when the microcontroller is idle and wake up via interrupts.
- ☐ C) Keep the microcontroller running at a low frequency to reduce power consumption.
- ☐ D) Use an external clock source to control power states of the microcontroller.

Expected Response

- ☐ A) Use a high-power microcontroller that completes tasks quickly to minimize active time.
- ☒ B) Implement deep sleep modes when the microcontroller is idle and wake up via interrupts.
- ☐ C) Keep the microcontroller running at a low frequency to reduce power consumption.
- ☐ D) Use an external clock source to control power states of the microcontroller.

Understanding GPIO register configuration

Problem Statement

What is the final value of my_register, given the following code snippet for the configuration?

```
// Initialize 16-bit register
my_register = 0x8000;
my_register |= (0x20 << 8);
```

Time Spent 1 min 11 secs
Difficulty Medium

Skills

Microcontrollers

Score

0 / 2

No Response

- ☐ 0x2000
- ☐ 0xA000
- ☐ 0x0000
- ☐ 0xC000

Expected Response

- ☐ 0x2000
- ☒ 0xA000
- ☐ 0x0000
- ☐ 0xC000

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Microcontroller Components

Problem Statement

A microcontroller typically contains several components. Which of the following is NOT typically included in a microcontroller?

Time Spent
0 min 17 secs

Difficulty
Medium

Skills

Basic Electronics

Score

2 / 2

Candidate Response

☐ A) Central Processing Unit (CPU)

☐ B) Memory (RAM and ROM)

☒ C) Graphics Processing Unit (GPU)

☐ D) Input/Output Ports

Expected Response

☐ A) Central Processing Unit (CPU)

☐ B) Memory (RAM and ROM)

☒ C) Graphics Processing Unit (GPU)

☐ D) Input/Output Ports

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

CPU Architecture

Problem Statement

The DMA differs from the interrupt mode by _____

Not applicable

Time Spent
1 min 0 sec

Difficulty
Easy

Skills

Embedded

Score

0 / 1

Candidate Response

☒ The involvement of the processor for the operation

☐ The method of accessing the I/O devices

☐ The amount of data transfer possible

☐ None of the mentioned

Expected Response

☐ The involvement of the processor for the operation

☐ The method of accessing the I/O devices

☐ The amount of data transfer possible

☒ None of the mentioned

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

CPU Architecture

Problem Statement

Why do most of the DSPs use Harvard architecture?

Not applicable

Time Spent Difficulty

0 min 25 secs

Easy

Skills

Embedded

Score

1/1

Candidate Response

☐ they provide greater bandwidth

☐ they provide more predictable bandwidth

☒ they provide greater bandwidth & also more predictable bandwidth

☐ none of the mentioned

Expected Response

☐ they provide greater bandwidth

☐ they provide more predictable bandwidth

☒ they provide greater bandwidth & also more predictable bandwidth

☐ none of the mentioned

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

CPU Architecture

Problem Statement

Traveo2 Cortex M0+ and M4 follow _____ and _____ architectures respectively

Not applicable

Time Spent Difficulty

0 min 38 secs

Easy

Skills

Embedded

Score

1/1

Candidate Response

☐ Harvard, Von Neumann

☒ Von Neumann, Harvard

☐ Harvard, Harvard

☐ Von Neumann, Von Neumann

Expected Response

☐ Harvard, Von Neumann

☒ Von Neumann, Harvard

☐ Harvard, Harvard

☐ Von Neumann, Von Neumann

- 123456789101112131415161718192021222324252627

Decimal to fixed-point

Problem Statement

How is -3.25 represented in Q3.13 format?

Not applicable

Time Spent

4 mins 39 secs

Difficulty

Medium

Skills

Number formats

Score

0 / 2

Candidate Response

☐ 0x9800

☐ 0xE800

☒ 0xA000

☐ 0x4000

Expected Response

☒ 0x9800

☐ 0xE800

☐ 0xA000

☐ 0x4000